PERFORMANCE TEST ONCENTRIFUGAL PUMP

AIM:-

To conduct the performance test on the given constant speed centrifugal pump and draw the following graphs.

1. Headv/s Discharge
2. Input Power v/s Discharge
3. Output Powerv/s Discharge
4. Efficiency v/s Discharge

INTRODUCTION

The centrifugal pump is widely used in house hold uses, oil Industries as well as different production, Chemical process Industries. The pump is used to lifts the liquid to higher level from the sump or water reservoir. The centrifugal pump has higher discharge and it is used under low head.

PRINCIPLE

Discharge, $$Q = C_d K\sqrt{2gh}\frac{m^3}{Sec}$$

C_d = Coefficient of discharge = 0.62

Constant, K $= \frac{a_1 a_2}{\sqrt{a_1^2 - a_2^2}}$

Where

a_1 = Area of pipe in m^2

a_2 = Area of orificem^2

Diameter of pipe, d_1 = 70 mm = 0.07 m

Diameter of orifice, d_2= 35 mm = 0.035 m

Acceleration due to gravity, g = 9.81 m/sec^2

Manometer difference, $h = \frac{(h_2 - h_1)\times 12.6}{100}$ m of water

Overall efficiency of pump, $= \dfrac{\text{OutputPower}}{\text{InputPower}} \times 100\ \%$

Output Power, OP $= \dfrac{\rho g Q H}{1000}$ kW

Where, $\rho =$ Density of Flowing liquid in kg/ m^3

$g =$ Acceleration due to gravity in m/sec^2

$Q =$ Discharge of flowing liquid in m^3/sec

H =Total head of pump in m

H =Total head in m = $H_d + H_s + l_d$
H_d= Delivery Pressure head in meters of water

H_s = Suction Pressure head in meters of water

l_d=Level difference between delivery and suction pressure headsin meters of water

Input power, IP $= \dfrac{n 3600 \eta_M \eta_T}{t E_m}$ kW

n = No of revolutions of energy meter

t = Time for 'n' rev of energy meter, in seconds.

Em = Energy meter constant rev/kW hour.

η_M = Motor efficiency = 80%

η_T=Transmission Efficiency = 90 %

PROCEDURE

1. The delivery valve, pressure gauge valve and vacuum gauge valve are fully closed.
2. Start the motor and open the delivery valve fully. Allow the flow to stabilise by regulatingthe gauges valves.
3. Note down the pressure gauge and vacuum gauge readings.
4. Note down the manometer readings h_1&h_2
5. Note down the time for 'n' revolution of the energy meter disc, and also note the speed of pump by using a tachometer.
6. The above procedure is repeated by gradually decreasing the flow of water by controlling the delivery valve (take minimum 6 sets of readings)

OBSERVATIONS

CENTRIFUGAL PUMP													
Sl.No	Speed of pump	Manometer readings			Pressure gauge Readings		Vacuum gauge Readings		Time for n revolutions of energy meter disc	Total Head	Discharge	Input IP	Output OP	Efficiency
		h_1	h_2	h= $(h_1 \sim h_2)$										
	rpm	cm	cm	m of water	Kgf/cm^2	mof water	mm of Hg	m of water	Sec	m	m^3/sec	kW	kW	%
1														
2														
3														
4														
5														
6														

SAMPLE CALCULATION(Set no: ....)

- **Discharge, Q** $= C_d K \sqrt{2gh}$ m^3/sec

Coefficient of discharge, Cd $= 0.62$

Constant, K $= \frac{a_1 a_2}{\sqrt{a_1^2 - a_2^2}}$

Diameter of pipe, d_1 = 70mm = 0.07 m

Area of pipe, a_1 = m^2

Diameter of Orifice, d_2 = 35 mm = 0.035 m

Area of Orifice, a_2 = m^2

Constant, K $= \frac{a_1 a_2}{\sqrt{a_1^2 - a_2^2}}$ = m^2

Left limb manometer reading, h_1 = cm of H_g

Right limb manometer reading, h2 = cm of H_g

Manometer difference, $h = \frac{(h_2 - h_1) \times 12.6}{100}$ m of water

Discharge, Q $= C_d \times K \sqrt{2gh} =$ = m^3/sec

- **Output Power, OP** $= \frac{\rho g Q H}{1000}\ kW$

Density of water, ρ = 1000 kg/m^3

Acceleration due to gravity, g = 9.81 m/sec^2.

Total head, $H = H_d + H_s + l_d$

H_d= Delivery Pressure head in meters of water

H_s = Suction Pressure head in meters of water

l_d=Level difference between delivery and suction pressure headsin meters of water

h_d - Delivery pressure head, (Kgf/cm^2)

h_s - Suction pressure head, (mm of Hg)

ρ_w- Density of water, 1000 kg/m^3

ρ_{Hg}-Density of Hg. 13600 kg/m^3

Delivery Pressure head $H_d = \frac{h_d}{\rho_w \times 10^{-4}} =$

Suction Pressure head $H_s = \frac{h_s \times 10^{-3} \times \rho_{Hg}}{\rho_w} =$

I_d $=$

Total Head, H in meters $=$

Output Power, OP $= \frac{\rho g Q H}{1000} =$ kW

- **Input power, IP** $=$

Input Power,IP $= \frac{n\, 3600 \eta_M \eta_T}{tE_m}$

No of Rev of Energy meter, n = 2 Rev.
Time for 2 Rev of Energy meter, t = sec
Motor Efficiency, η_M = 80%
Transmission Efficiency, η_T = 90 %

Energy meter Constant, E_m = 75 Rev/ kW hour.

Input power, IP $= \frac{n\, 3600 \eta_M \eta_T}{tE_m} =$ = kW

Overall Efficiency, η_O $= \frac{OP}{IP} \times 100 =$ = %

GRAPHS:

1. Head v/s Discharge
2. Input Power v/s Discharge
3. Output Power v/s Discharge
4. Efficiency v/s Discharge

RESULT:

The required graphs are plotted.

INFERENCE: